

Self-Reflective Report

Java Programming Course

1. Introduction

This report reflects on my 10-week learning journey in Java programming. Starting as a beginner with limited programming experience, I gradually built a solid foundation in Java syntax, object-oriented programming (OOP), and core development skills through weekly tutorials and practical exercises.

2. Knowledge Gained from Each Tutorial

- Week 1 – Java Fundamentals: Learned Java setup, wrote my first "Hello World" program, understood primitive data types (int, double, char, boolean), variables, identifiers, and basic syntax.
- Week 2 – Input/Output & Operators: Mastered Scanner for user input, System.out.println() for output, arithmetic and relational operators, and type casting.
- Week 3-4 – Control Structures: Implemented conditional statements (if-else, switch) and loops (for, while, do-while) to control program flow.
- Week 5 – Methods: Learned to declare methods with parameters and return types, understood variable scope, and practiced method overloading.
- Week 6 – Classes & Objects: Grasped core OOP concepts – creating classes with attributes and methods, using constructors, and implementing encapsulation.

- Week 7 – Inheritance & Polymorphism: Explored inheritance hierarchies, method overriding, abstract classes, interfaces, and polymorphic behavior.
- Week 8-9 – Arrays & ArrayLists: Worked with single and multi-dimensional arrays, learned ArrayList for dynamic collections, and practiced searching and sorting.
- Week 10 – Exception Handling & File I/O: Implemented try-catch blocks for robust error handling and used FileReader/FileWriter for file operations.

3. Challenges Encountered

- Understanding the difference between primitive types and reference types.
- Grasping inheritance and polymorphism concepts.
- Debugging logical errors in loops and conditionals.
- Managing exceptions and file I/O operations correctly.

4. How Challenges Were Overcome

- Practiced with memory diagrams to visualize primitives vs. objects.
- Wrote small test programs to experiment with inheritance and polymorphism.
- Used IDE debuggers (breakpoints, step-through) to trace program execution.
- Referenced Java documentation and online resources for exception handling.

5. Improvements in Java Programming Skills

- Before: Basic syntax knowledge, difficulty with error messages.
- After: Confident in writing structured, object-oriented Java code; proficient in using IDEs and debuggers; able to read and understand Java documentation.

6. Understanding of Object-Oriented Programming Concepts

I now understand the four pillars of OOP:

- Encapsulation: Hiding internal data using private modifiers and providing controlled access via getters/setters.
- Inheritance: Creating hierarchical class relationships using extends.
- Polymorphism: Using a single interface to represent different underlying forms (method overriding, interfaces).
- Abstraction: Using abstract classes and interfaces to define contracts without implementation details.

7. Future Learning Plans

- Explore Java Collections Framework (HashMap, HashSet, etc.).
- Learn GUI development with JavaFX or Swing.
- Study multithreading and concurrency.

- Build a complete Java project (e.g., Library Management System).

8. Conclusion

This 10-week Java course has been transformative. I progressed from writing simple console programs to developing object-oriented applications with robust error handling. The structured weekly tutorials, combined with hands-on coding, built my confidence and competence in Java programming. I look forward to applying these skills in future projects and continuing my learning journey.